Noisy Coding in Decision-Making under Risk:

Bridging Description and Experience

Jilong Wu

Risl$\alpha\beta$, Gent University

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GHENT UNIVERSITY

RISLαβ

RISK, INSURANCE, AND SAVINGS LABORATORY

Background – Two Paradigms

The *description* paradigm:

- All lottery information given explicitly:

Please Make Your Choice

20% winning $100 80% winning nothing	100% winning $20

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The *experience* paradigm (less explored):

- To sample outcomes before choice

Please Sample Options before Choose

Option A

Option B

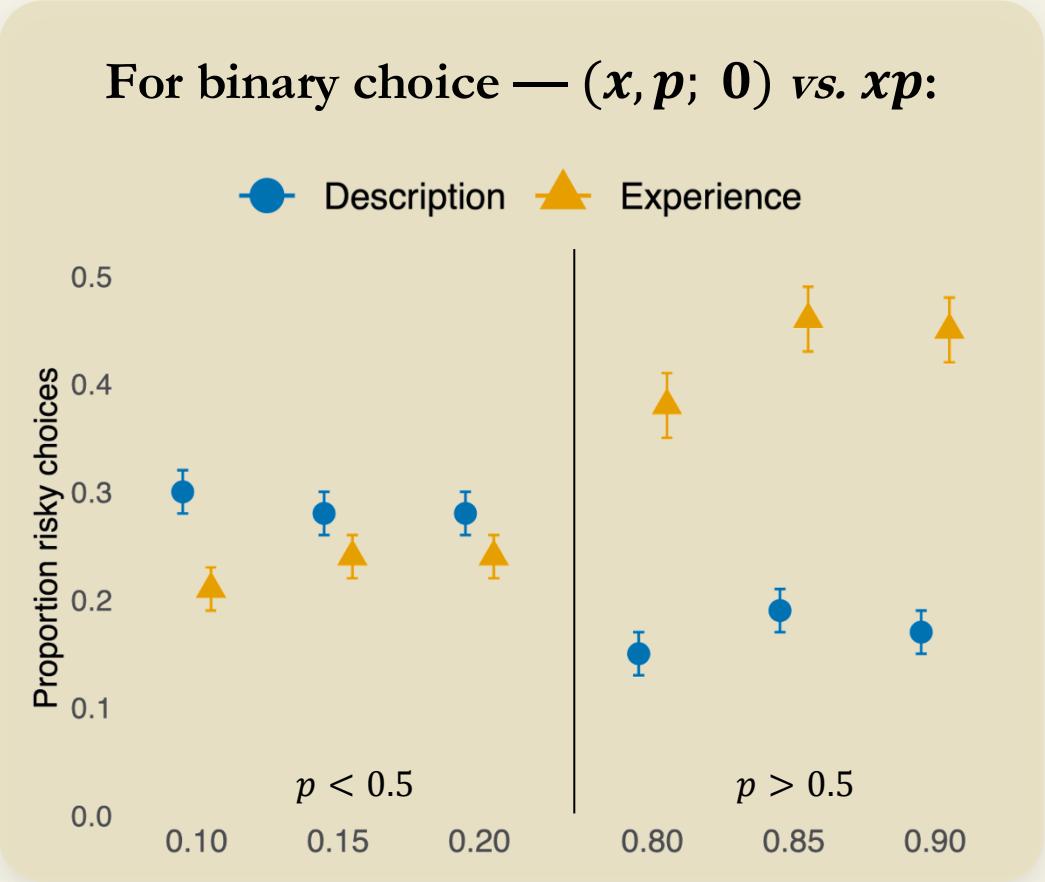
→

Please Make Your Choice

Option A

Option B

Background – The Gap

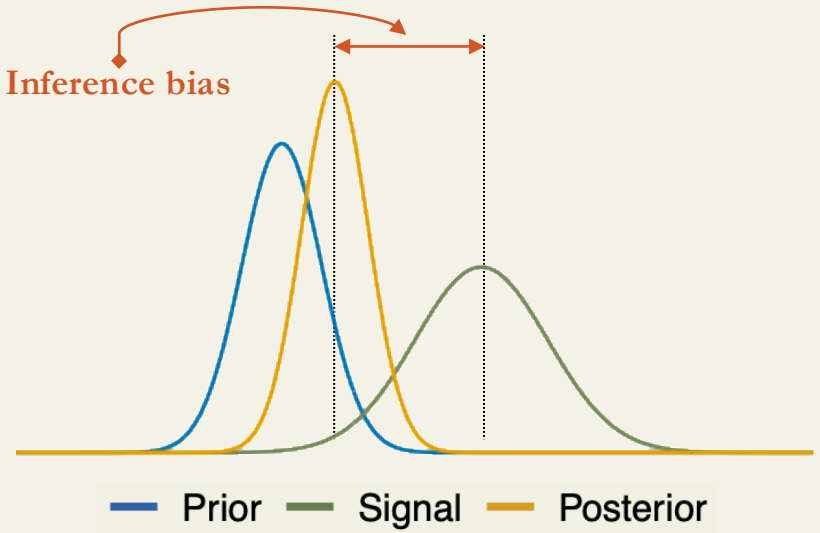


Source: Figure 1 in Oprea & Vieider (2026).

The description–experience gap

- People make very different choices under these two paradigms.
- No clear explanation by standard models of decision-making under risk.

Noisy Coding Model



Noisy coding model

Zhang et al. (2020, PNAS); Khaw et al. (2021, REStud); Frydman and Jin (2022, QJE); Enke and Graeber (2023, QJE); Oprea (2024, AER); Vieider (2024, Manag. Sci.):

- **A unified framework (same cognitive mechanism) for both paradigms** (Oprea and Vieider, 2026).
- **Inference bias helps explain many choice patterns.**
- **Sampling bias additionally arises under *experience*.**

Work packages

The Allais Paradox

Under *description:*

$$(4, 0.8) \prec (3, 1)$$

p multipled by $\frac{1}{10}$ $\updownarrow$

$$(4, 0.08) \succ (3, 0.1)$$

Under *experience*: ?

NCM predicts a reversed pattern.

The Stake Effects

Under *description:*

$$(10, 0.2) \succ (2, 1)$$

x multipled by 40 $\updownarrow$

$$(400, 0.2) \prec (80, 1)$$

Under *experience*: ?

NCM predicts a reversed pattern.

Mental Sampling

Most challenging:
Operationalize the mental number representation under *description.*

External collaboration with Prof. Zhu.
Machine-Augmented Cognition Lab,
University of Hong Kong

Summary

Theoretical & Empirical Contribution

- Model extension
- Behavior under *experience*
- Discussion about cognitive foundations of risky choice

Grounded Policy Implications

- Risk communication
- Decision aids
- Welfare justification

Appendix – I am the right person

Why:

- **Very familiar with the research topic**:
  - A large scale meta study (under-review: the Economic Journal).
  - Several working papers on noisy coding and cognitive friction.
- **Independent and active researcher**: proficient skills; conduct more than 20 experiments; 10 invited seminar talks and many conference talks.
- **A strong international network**:
  - Prof. Ferdinand Vieider, UGent
  - Prof. Ryan Opera, UC Berkeley
  - Prof. Songfa Zhong, NUSingapore
  - Prof. Savadori Nunnari, Bocconi
  - Prof. Robert Lensink, UGroningen
  - Prof. Taisuke Imai, UOsaka
  - Prof. Jianying Qiu, URadboud.
  - Prof. Jianqiao Zhu, UHongKong
  - etc.

Appendix – Experiment setup

Binary choices between lotteries (bags) are the basic decision unit:

- We represent probabilities as frequencies—the number of coins in each amount.
- *Description* Example: (10, 0.2) *vs.* 2:

Please choose the bag you prefer:

Bag A
8 coins are worth $10
2 coins are worth $0

Bag B
10 coins are worth $2

Appendix – Experiment setup

Two sample-setup examples (*experience*):

- The free sampling treatment:

Please sample the bags:

Bag A **Bag B**

The drew coin is worth **$10**

- The complete sampling treatment (no replacement):

Please sample the bags:

Bag A **Bag B**

3/10: The drew coin is worth **$10**

Appendix – Experiment setup

The mental sampling example (paced random generation method):

Please imagine drawing balls from Bag A, and click the corresponding button at a time.

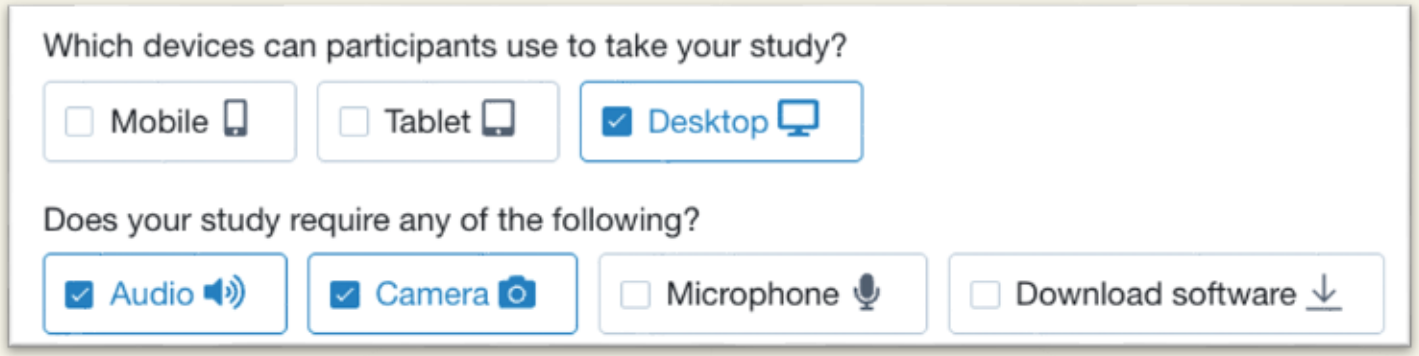
Bag A
8 coins are worth $10
2 coins are worth $0

Bag B
10 coins are worth $2

$10 $0

Experiment Management

- To monitor the online experiment and ensure engagement, invite subjects to enter the zoom room with camera on:



Prolific

- Detailed preregistration (Open Science Framework): e.g., JPE Micro; Experi. Econ.
- Pilot study to refine the experimental design and inform power analysis.
- Access to external labs (e.g., University of Groningen, Radboud University, Maastricht University)